

Q1. Applet Life Cycle

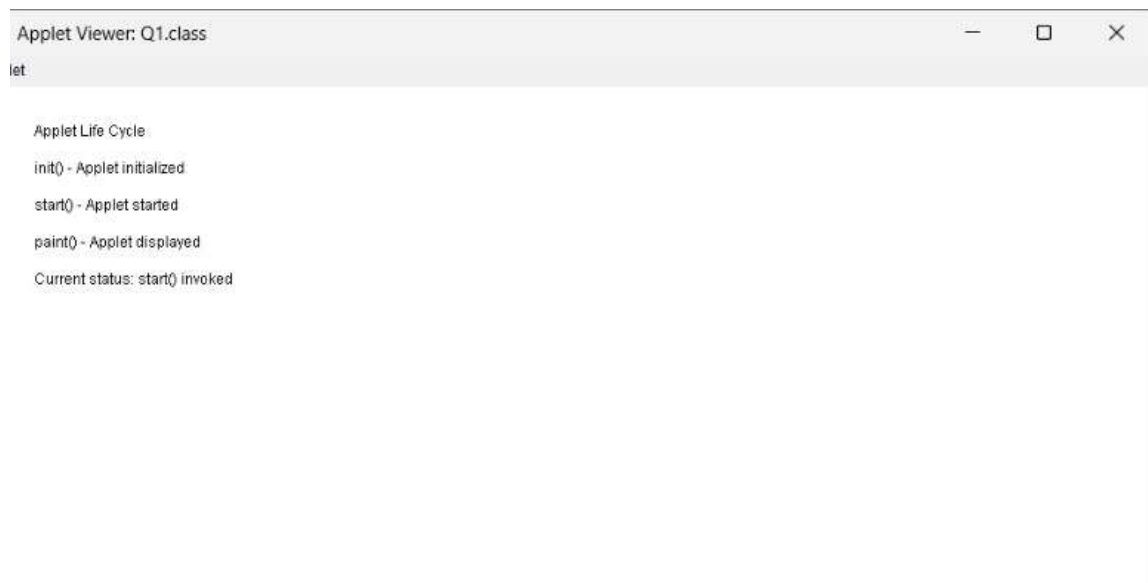
```
import java.applet.Applet;
import java.awt.*;

public class LifeCycle extends Applet {
    String msg = "";
    public void init() {
        msg = "init() called";
    }
    public void start() {
        msg = msg + "\nstart() called";
        repaint();
    }
    public void paint(Graphics g) {
        g.drawString(msg, 50, 80);
        g.drawString("paint() called", 50, 110);
    }
    public void stop() {
        msg = msg + "\nstop() called";
    }
    public void destroy() {
        System.out.println("destroy() called");
    }
}
```

HTML:

```
<html>
```

```
<body>
<applet code="LifeCycle.class" width="400" height="200">
</applet>
</body>
</html>
```



Q2. Drawing House, Sun, Trees, Road and Clouds

```
import java.applet.Applet;
import java.awt.*;
public class Scene extends Applet {
    public void paint(Graphics g) {
        // Sky
        g.setColor(Color.cyan);
        g.fillRect(0, 0, 500, 300);
        // Sun
        g.setColor(Color.yellow);
```

```
g.fillOval(400, 30, 60, 60);
// Clouds
g.setColor(Color.white);
g.fillOval(60, 50, 70, 40);
g.fillOval(100, 40, 70, 50);
g.fillOval(140, 50, 70, 40);
// Ground
g.setColor(Color.green);
g.fillRect(0, 250, 500, 150);
// Road
g.setColor(Color.darkGray);
g.fillRect(200, 250, 100, 150);
// House body
g.setColor(Color.orange);
g.fillRect(100, 160, 180, 100);
// Roof
g.setColor(Color.red);
Polygon roof = new Polygon();
roof.addPoint(80, 160);
roof.addPoint(190, 80);
roof.addPoint(300, 160);
g.fillPolygon(roof);
// Door
g.setColor(Color.black);
g.fillRect(170, 200, 40, 60);
// Window
g.setColor(Color.white);
```

```

g.fillRect(120, 180, 40, 40);
// Tree 1
g.setColor(Color.green);
g.fillOval(20, 170, 70, 80);
g.setColor(Color.black);
g.fillRect(50, 240, 15, 60);
// Tree 2
g.setColor(Color.green);
g.fillOval(350, 170, 70, 80);
g.setColor(Color.black);
g.fillRect(380, 240, 15, 60);
}
}

```

HTML:

```

<html>
<body>
<applet code="Scene.class" width="500" height="400">
</applet>
</body>
</html>

```



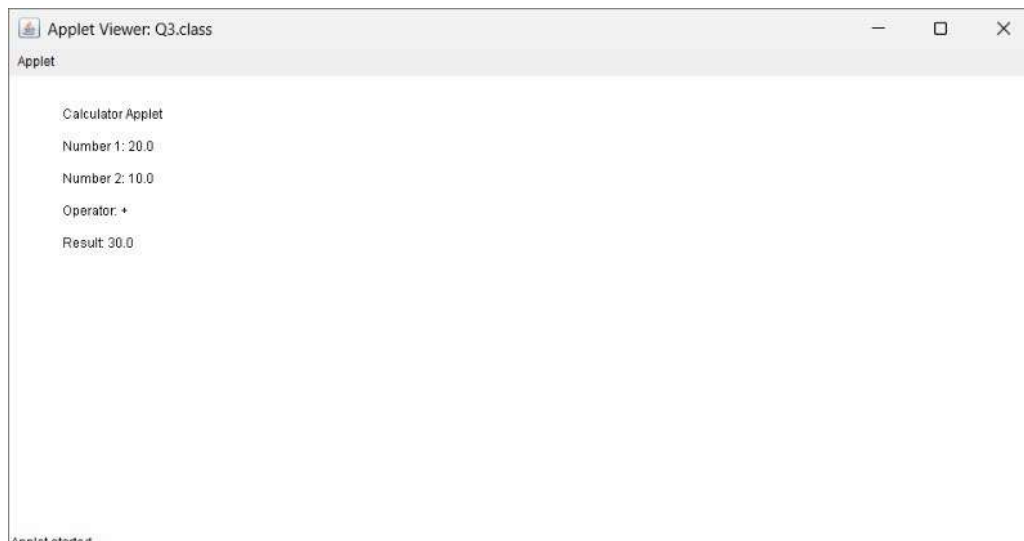
Q3. Calculator Applet Using Parameters

```
import java.applet.Applet;
import java.awt.*;

public class CalculatorApplet extends Applet {
    public void paint(Graphics g) {
        double a = Double.parseDouble(getParameter("num1"));
        double b = Double.parseDouble(getParameter("num2"));
        String op = getParameter("op");
        double result = 0;
        if (op.equals("+"))
            result = a + b;
        else if (op.equals("-"))
            result = a - b;
        else if (op.equals("*"))
            result = a * b;
        else if (op.equals("/"))
            result = a / b;
        g.drawString("First Number : " + a, 50, 60);
        g.drawString("Second Number : " + b, 50, 80);
        g.drawString("Operator : " + op, 50, 100);
        g.drawString("Result : " + result, 50, 120);
    }
}
```

HTML:

```
<html>
<body>
<applet code="CalculatorApplet.class" width="400" height="200">
<param name="num1" value="20">
<param name="num2" value="5">
<param name="op" value="+">
</applet>
</body>
</html>
```



Q4. Student Profile Using AWT Components

```
import java.awt.*;
import java.awt.event.*;
public class StudentProfile extends Frame implements ActionListener {
    Label nameLabel, regLabel, genderLabel, deptLabel, yearLabel, skillLabel;
    TextField name, reg;
    Checkbox male, female;
```

```
CheckboxGroup gender;
Choice dept, year;
List skills;
TextArea result;
Button generate;
StudentProfile() {
    setTitle("Student Profile");
    setSize(500, 500);
    setLayout(new FlowLayout());
    nameLabel = new Label("Name");
    name = new TextField(20);
    regLabel = new Label("Register Number");
    reg = new TextField(20);

    genderLabel = new Label("Gender");
    gender = new CheckboxGroup();
    male = new Checkbox("Male", gender, false);
    female = new Checkbox("Female", gender, false);
    deptLabel = new Label("Department");
    dept = new Choice();
    dept.add("CSE");
    dept.add("ECE");
    dept.add("AIDS");
    dept.add("EEE");
    yearLabel = new Label("Year");
    year = new Choice();
    year.add("1st Year");
```

```
year.add("2nd Year");
year.add("3rd Year");
year.add("4th Year");
skillLabel = new Label("Skills");
skills = new List(4, true);
skills.add("Java");
skills.add("Python");
skills.add("C");
skills.add("HTML");
generate = new Button("Generate Report");
result = new TextArea(8, 40);
add(nameLabel);
add(name);
```

```
add(regLabel);
add(reg);
add(genderLabel);
add(male);
add(female);
add(deptLabel);
add(dept);
add(yearLabel);
add(year);
add(skillLabel);
add(skills);
add(generate);
add(result);
```



```

generate.addActionListener(this);
addWindowListener(new WindowAdapter() {
    public void windowClosing(WindowEvent e) {
        dispose();
    }
});
setVisible(true);
}

public void actionPerformed(ActionEvent e) {
    String g = gender.getSelectedCheckbox().getLabel();
    String s = "";
    String selected[] = skills.getSelectedItems();
    for (String x : selected)
        s = s + x + " ";

    result.setText(
        "Name: " + name.getText() +
        "\nRegister Number: " + reg.getText() +
        "\nGender: " + g +
        "\nDepartment: " + dept.getSelectedItem() +
        "\nYear: " + year.getSelectedItem() +
        "\nSkills: " + s
    );
}

public static void main(String args[]) {
    new StudentProfile();
}

```

}

Student Profile

Name: Madhu

Register Number: 123

Gender: ☐ Male ☒ Female

Department: CSE

Year: 1st Year

Skills: ☒ Java ☐ Python ☐ HTML

Programming Language: Java, Python, C, C++

Generate Report

STUDENT PROFILE
Name: Madhu
Register Number: 123
Gender: Female

Q5. GridLayout Calculator

```
import java.awt.*;
import java.awt.event.*;

public class GridCalculator extends Frame implements ActionListener {
    TextField t;
    double num1, num2;
    char op;

    GridCalculator() {
        setTitle("Calculator");
        setSize(300, 400);

        t = new TextField();
        add(t, BorderLayout.NORTH);

        Panel p = new Panel();
        p.setLayout(new GridLayout(4, 4));
```

```

String buttons[] = {
    "7", "8", "9", "+",
    "4", "5", "6", "-",
    "1", "2", "3", "*",
    "0", "=", "C", "/"
};

for (String b : buttons) {
    Button btn = new Button(b);
    btn.addActionListener(this);
    p.add(btn);
}

add(p, BorderLayout.CENTER);
addWindowListener(new WindowAdapter() {
    public void windowClosing(WindowEvent e) {
        dispose();
    }
});

setVisible(true);
}

public void actionPerformed(ActionEvent e) {
    String s = e.getActionCommand();
    if (s.equals("C")) {
        t.setText("");
    }
    else if (s.equals("+") || s.equals("-") ||
        s.equals("*") || s.equals("/")) {
        num1 = Double.parseDouble(t.getText());
    }
}

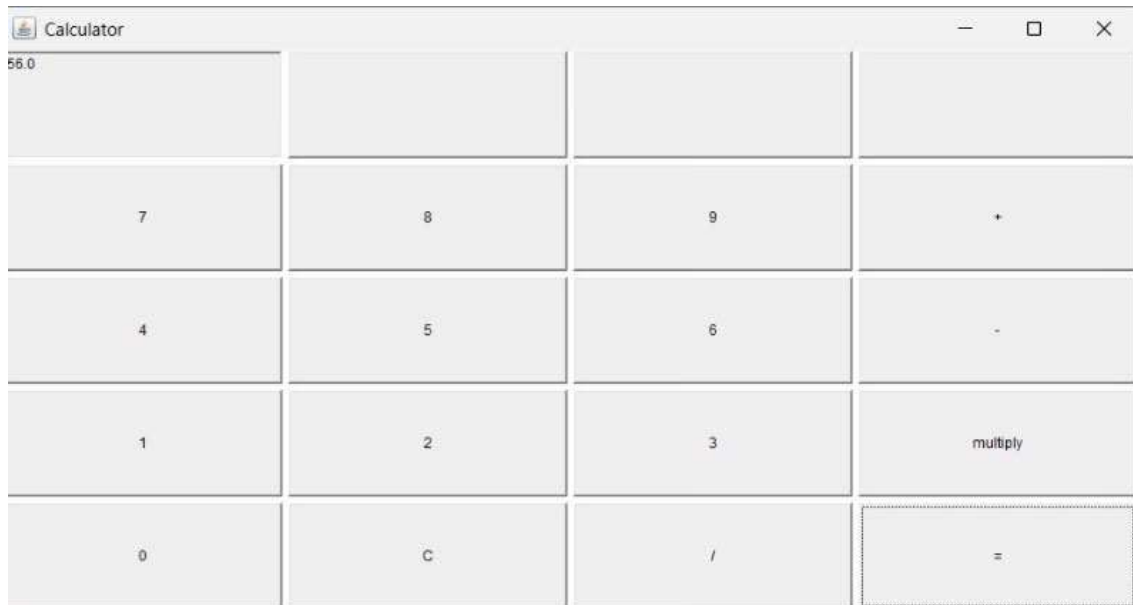
```

```

        op = s.charAt(0);
        t.setText("");
    }
    else if (s.equals("=")) {
        num2 = Double.parseDouble(t.getText());
        double result = 0;
        if (op == '+')
            result = num1 + num2;
        else if (op == '-')
            result = num1 - num2;
        else if (op == '*')
            result = num1 * num2;
        else if (op == '/')
            result = num1 / num2;
        t.setText("" + result);
    }
    else {
        t.setText(t.getText() + s);
    }
}

public static void main(String args[]) {
    new GridCalculator();
}
}

```



Q6. GridBagLayout Student Registration

```
import java.awt.*;
import java.awt.event.*;
public class Registration extends Frame implements ActionListener {
    TextField name, reg, email;
    Choice dept, year;
    Checkbox male, female;
    CheckboxGroup gender;
    Button submit, reset;
    Registration() {
        setTitle("Student Registration");
        setSize(500, 400);
        setLayout(new GridBagLayout());
        GridBagConstraints g = new GridBagConstraints();
        g.insets = new Insets(5, 5, 5, 5);
        name = new TextField(20);
```

```
reg = new TextField(20);
email = new TextField(20);
dept = new Choice();
dept.add("CSE");
dept.add("ECE");
dept.add("AIDS");
year = new Choice();
year.add("1st Year");
year.add("2nd Year");
year.add("3rd Year");
year.add("4th Year");
gender = new CheckboxGroup();
male = new Checkbox("Male", gender, false);
female = new Checkbox("Female", gender, false);
submit = new Button("Submit");
reset = new Button("Reset");
addComponent(new Label("Name"), name, 0, g);
addComponent(new Label("Register Number"), reg, 1, g);
addComponent(new Label("Department"), dept, 2, g);
addComponent(new Label("Email"), email, 3, g);
g.gridx = 0;
g.gridy = 4;
add(new Label("Gender"), g);
g.gridx = 1;
add(male, g);
g.gridx = 2;
add(female, g);
```

```

g.gridx = 0;
g.gridy = 5;
add(new Label("Year"), g);
g.gridx = 1;
add(year, g);
g.gridx = 0;
g.gridy = 6;
add(submit, g);

g.gridx = 1;
add(reset, g);
submit.addActionListener(this);
reset.addActionListener(this);
addWindowListener(new WindowAdapter() {
    public void windowClosing(WindowEvent e) {
        dispose();
    }
});
setVisible(true);
}

void addComponent(Component label, Component field,
                  int row, GridBagConstraints g) {
    g.gridx = 0;
    g.gridy = row;
    add(label, g);
    g.gridx = 1;
    g.gridwidth = 2;

```

```

        add(field, g);
        g.gridwidth = 1;
    }
    public void actionPerformed(ActionEvent e) {
        if (e.getSource() == submit) {
            String genderValue =
                gender.getSelectedCheckbox() == null ?
                "" : gender.getSelectedCheckbox().getLabel();

            System.out.println("Name: " + name.getText());
            System.out.println("Register Number: " + reg.getText());
            System.out.println("Department: " + dept.getSelectedItem());
            System.out.println("Email: " + email.getText());
            System.out.println("Gender: " + genderValue);
            System.out.println("Year: " + year.getSelectedItem());
        }
        if (e.getSource() == reset) {
            name.setText("");
            reg.setText("");
            email.setText("");
        }
    }
    public static void main(String args[]) {
        new Registration();
    }
}

```


Q7. BorderLayout with Event Handling

```
import java.awt.*;
import java.awt.event.*;

public class BorderDemo extends Frame implements ActionListener {
    Label center;

    BorderDemo() {
        setTitle("BorderLayout Demo");
        setSize(500, 400);

        setLayout(new BorderLayout());
        Label header = new Label("HEADER", Label.CENTER);
        center = new Label("Display Area", Label.CENTER);
        Button b1 = new Button("Home");
        Button b2 = new Button("Menu");
        Button b3 = new Button("Options");
        Button b4 = new Button("Exit");
        add(header, BorderLayout.NORTH);
        add(b1, BorderLayout.WEST);
        add(b2, BorderLayout.SOUTH);
        add(b3, BorderLayout.EAST);
        add(center, BorderLayout.CENTER);
        b1.addActionListener(this);
        b2.addActionListener(this);
        b3.addActionListener(this);
        b4.addActionListener(this);
        addWindowListener(new WindowAdapter() {
            public void windowClosing(WindowEvent e) {
```

```

        dispose();
    }
});
setVisible(true);
}

public void actionPerformed(ActionEvent e) {
    center.setText("You clicked: " +
        e.getActionCommand());

    public static void main(String args[]) {
        new BorderDemo();
    }
}

```

Q8. CardLayout with Event Handling

```

import java.awt.*;
import java.awt.event.*;
public class CardDemo extends Frame implements ActionListener {
    CardLayout card;
    Panel panel;
    Button next, previous, submit;
    CardDemo() {
        setTitle("Student Registration");
        setSize(500, 350);
        card = new CardLayout();
        panel = new Panel(card);
    }
}

```

```
// Card 1
Panel p1 = new Panel();
p1.add(new Label("Student Information"));
p1.add(new Label("Name:"));
p1.add(new TextField(15));

// Card 2
Panel p2 = new Panel();
p2.add(new Label("Academic Information"));
p2.add(new Label("Department:"));
p2.add(new TextField(15));

// Card 3
Panel p3 = new Panel();
p3.add(new Label("Confirmation / Report"));
p3.add(new Label("Registration Completed"));
panel.add(p1, "card1");
panel.add(p2, "card2");
panel.add(p3, "card3");
add(panel, BorderLayout.CENTER);

Panel buttons = new Panel();
previous = new Button("Previous");
next = new Button("Next");
submit = new Button("Submit");
buttons.add(previous);
buttons.add(next);
buttons.add(submit);
add(buttons, BorderLayout.SOUTH);
previous.addActionListener(this);
```

```

        next.addActionListener(this);
        submit.addActionListener(this);
        addWindowListener(new WindowAdapter() {
            public void windowClosing(WindowEvent e) {
                dispose();
            }
        });

        setVisible(true);
    }

    public void actionPerformed(ActionEvent e) {
        String s = e.getActionCommand();
        if (s.equals("Next"))
            card.next(panel);
        else if (s.equals("Previous"))
            card.previous(panel);
        else if (s.equals("Submit"))
            card.show(panel, "card3");
    }

    public static void main(String args[]) {
        new CardDemo();
    }
}

```